



THE AMERICAN
BOARD OF SURGERY

Using LLMs to Score ABS Oral Certifying Exams: A Pilot Study at The American Board of Surgery

AMERICAN BOARD OF SURGERY

Nothing to Disclose

Overview

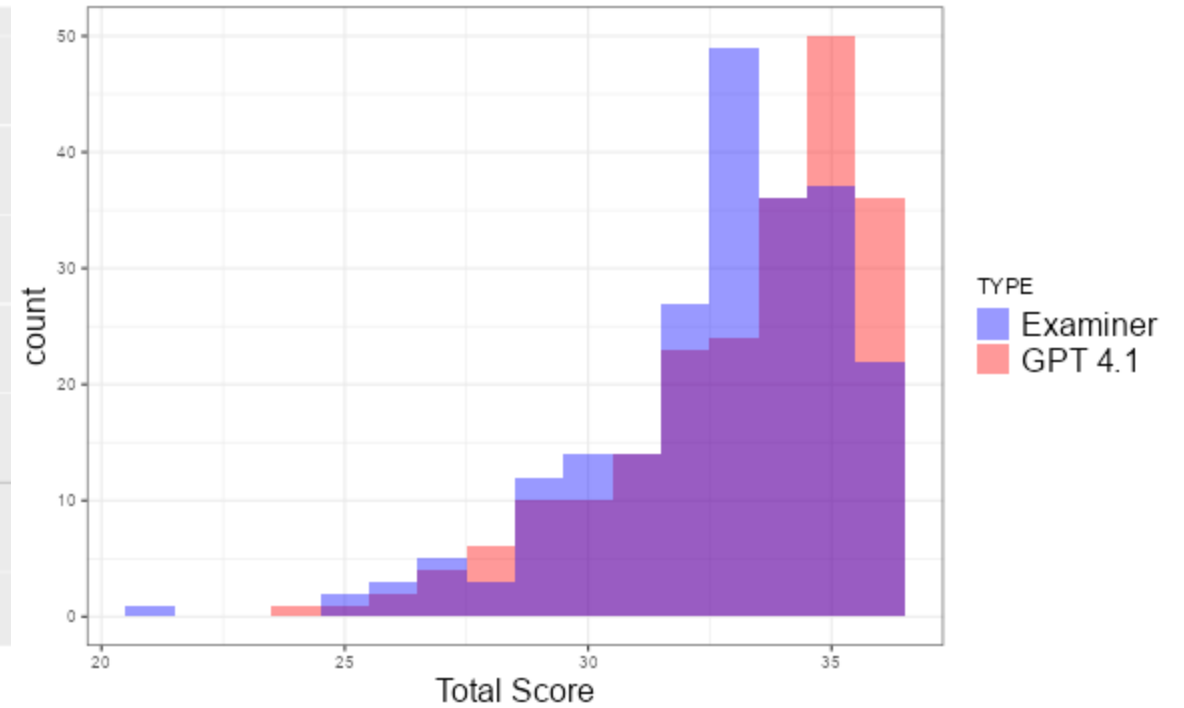
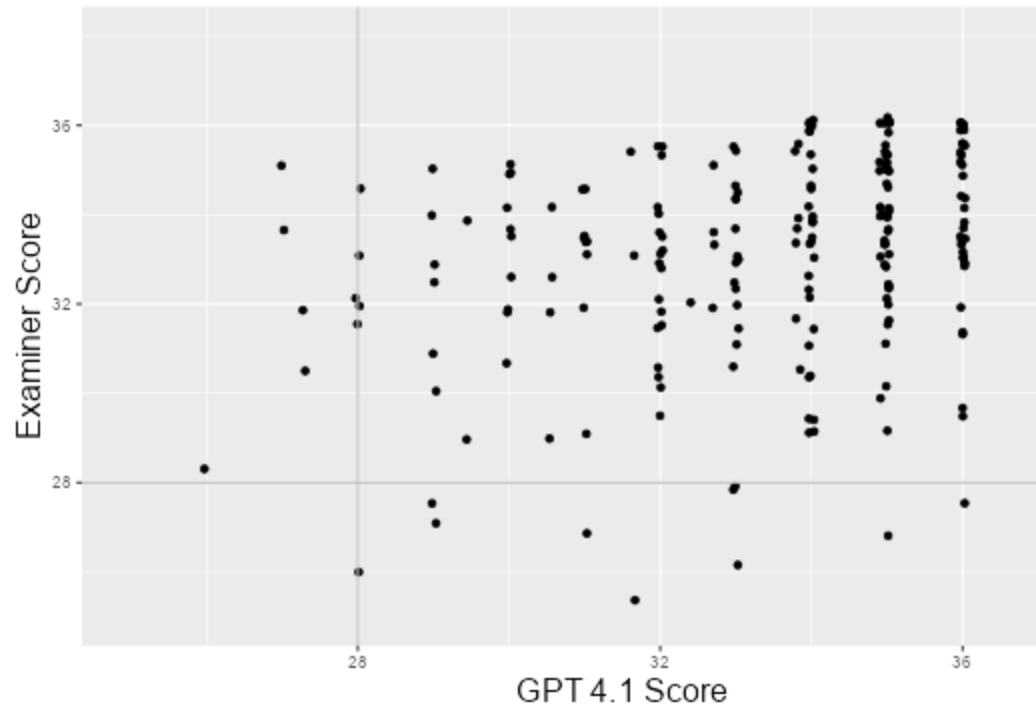
Technical & Security Configurations

Model Training & Results for Scoring

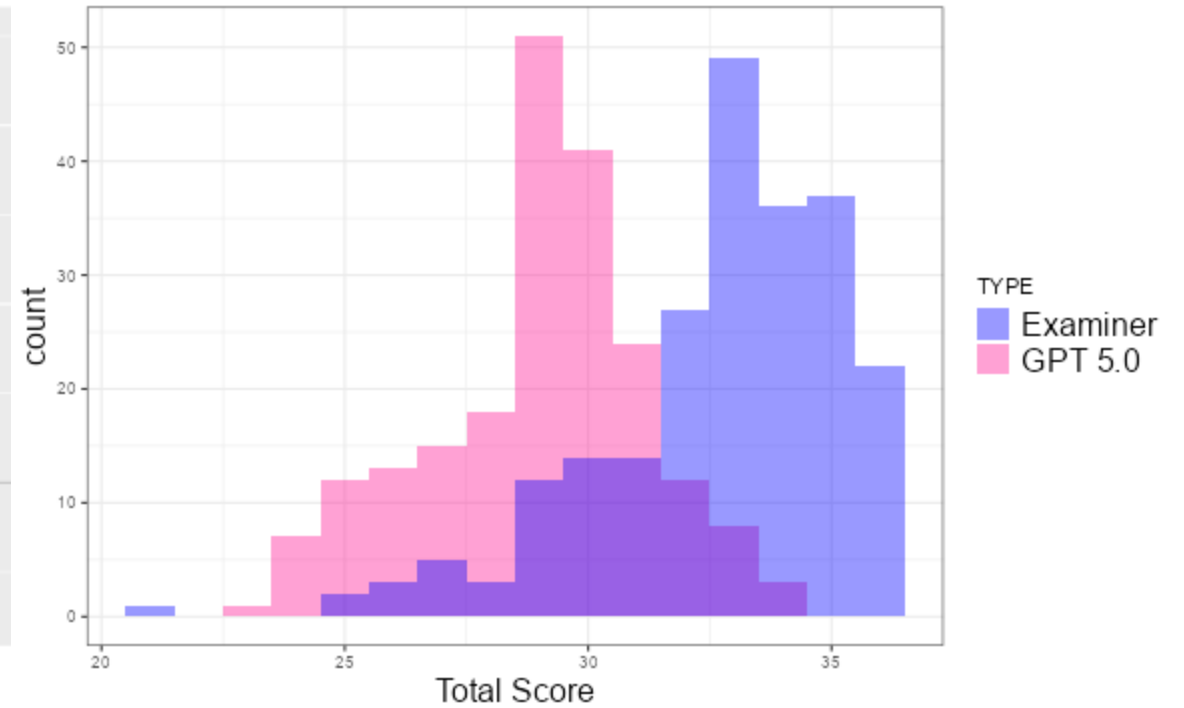
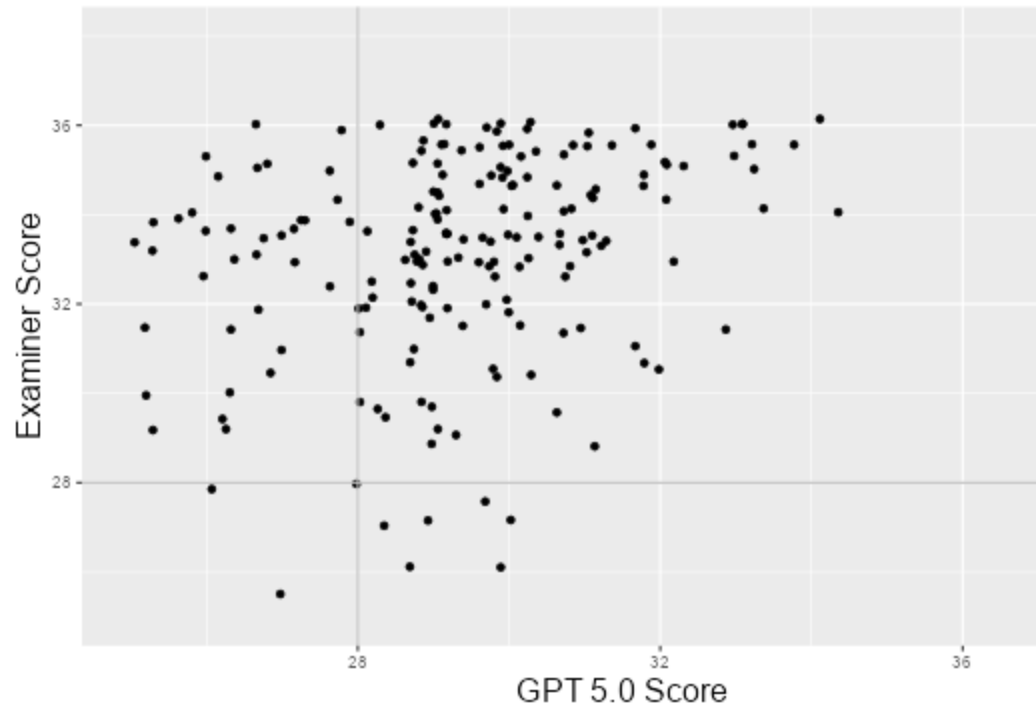
Scoring

- Four agents: GPT 4.1, GPT 5.0, Claude Opus, and Claude Sonnet
- Inputs
 - Examiner training materials
 - Case content
 - De-identified exam transcripts
 - Same prompt and agent instructions across all four agents
- AI-generated scores were compared to the average of two examiner scores for each case

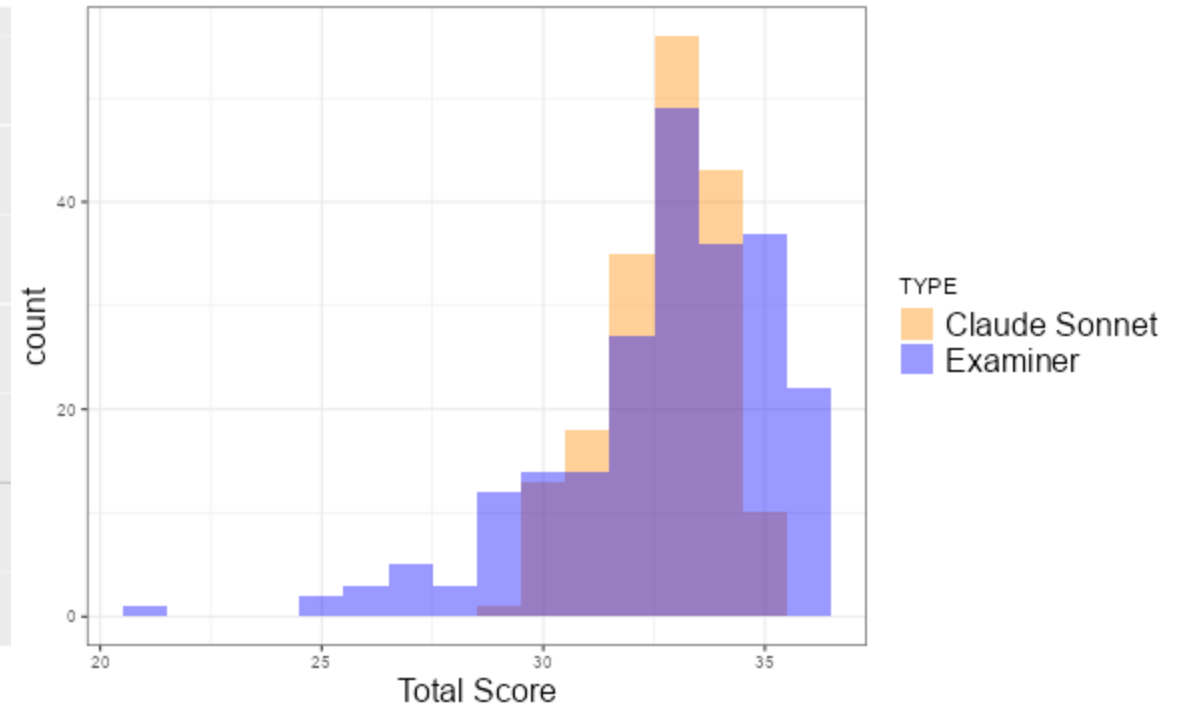
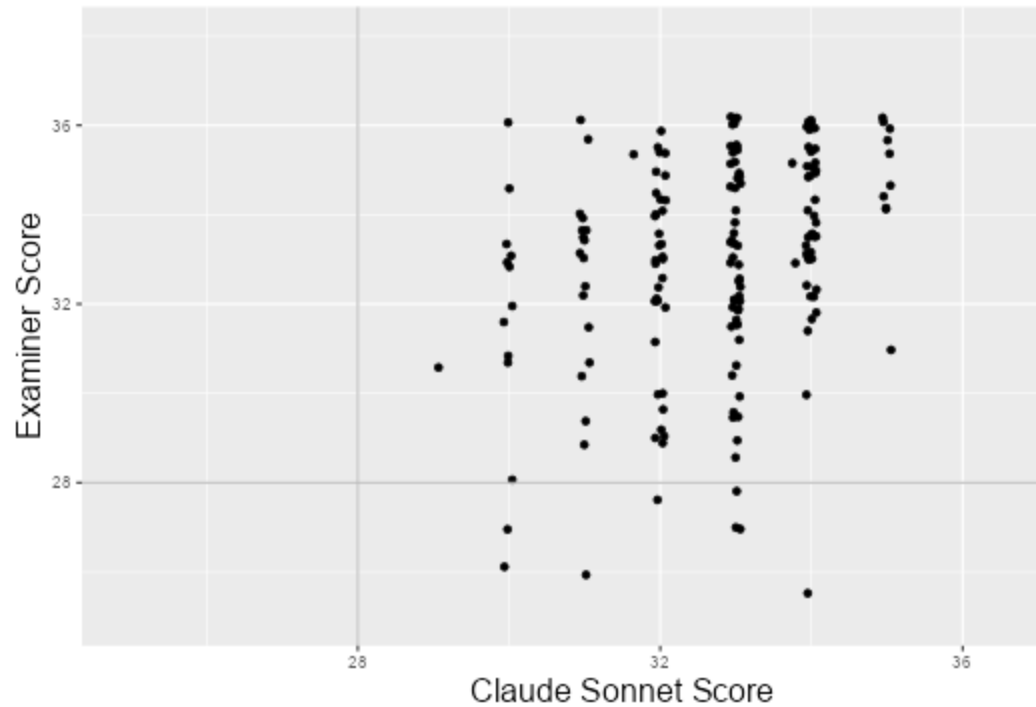
GPT 4.0 ($R^2 = .12$)



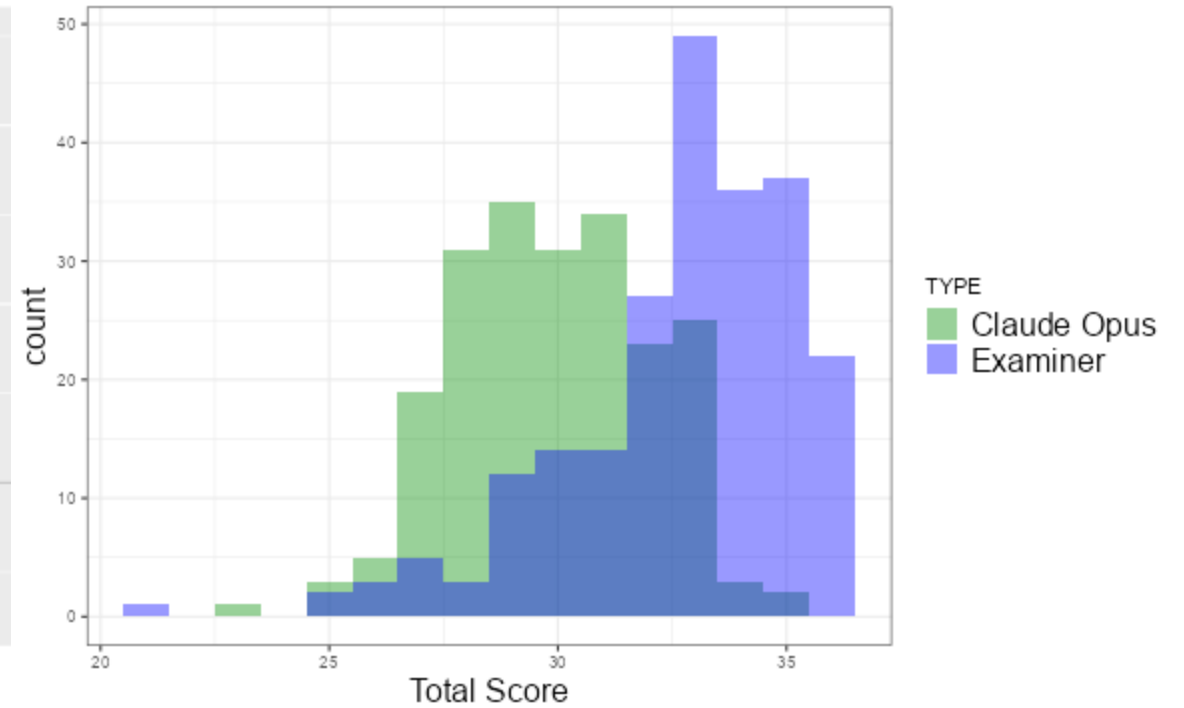
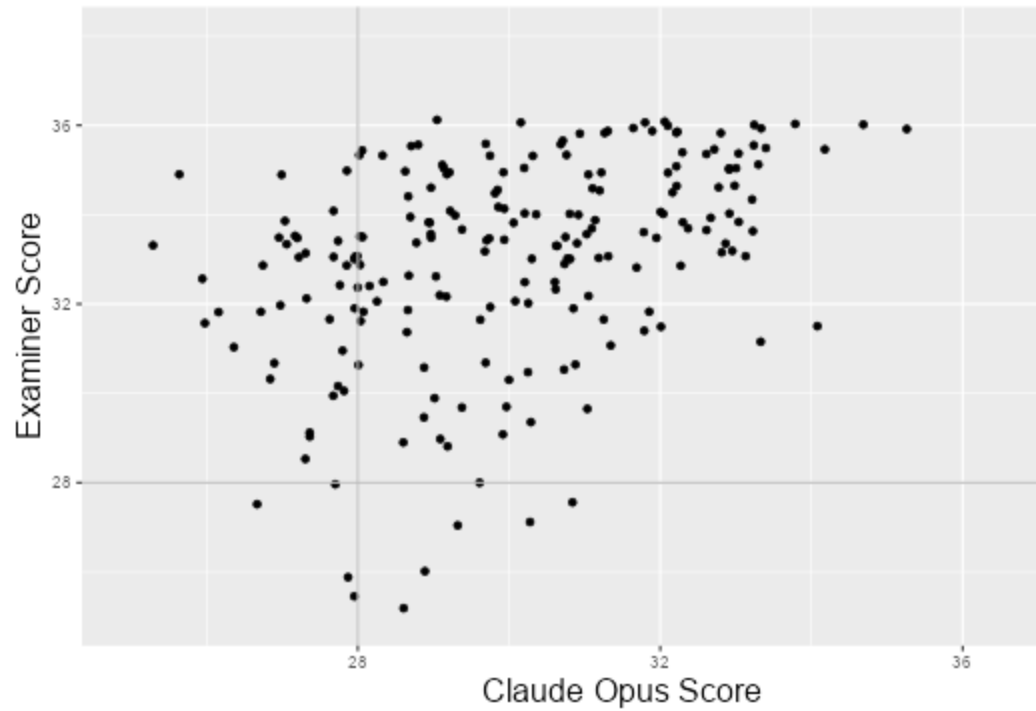
GPT 5.1 ($R^2 = .10$)



Claude Sonnet ($R^2 = .10$)



Claude Opus ($R^2 = .18$)



Overall Patterns

- AI scores tended to be lower than human scores
- Correlations were low
- All models showed large proportions of examinees with different pass/fail statuses for human vs. AI raters

AI Scoring: Issues

- Response quality
 - Hallucinated cases (e.g., Case V0) and duplicate case scores
 - Difficulty obtaining failing case scores for some models
 - Refusal to respond
- Prompt development
 - Erring on the side of patient safety
 - Reminders of score definitions
 - Optimal prompts may differ across models, complicating comparisons
- Logistics
 - Differences in output formatting despite clear prompting instructions
 - Handling response refusals and token limits
 - Infrastructure for training materials and data organization

Model Training & Results for Examiner Feedback

Exploring Utility of AI for Examiner Feedback

- Current Feedback Mechanisms
 - Examiner feedback reports of scoring behaviors
 - Co-examiner evaluation forms
 - Informal reports of examiner behavior
- Limitations of Current Mechanisms
 - Overwhelmingly positive
 - Limited to scoring behaviors (harshness/leniency, inconsistency)
 - No structured feedback on *conduct of the exam*
- Can LLMs augment our existing feedback?

Approach

- Inputs (similar to inputs for scoring)
 - Examiner training materials
 - Case content
 - Deidentified exam transcripts
- Prompt Development
 - Request feedback on conduct of the exam
 - Reading the stem to candidate & adherence to case content
 - Time management
 - Ability to probe/request more information; open questioning versus leading questions; no verbal signals on performance
 - Provided neutral exam issues
 - Behavioral issues
- Identify specific examiners to test
 - Overly lenient/harsh; new/experienced examiners
 - Known issues (e.g., did not read the stem, reports of behavioral issues)
 - Issues reported via co-examiner feedback form

Challenges Encountered

- Feedback given for incorrect examiner
 - Each examiner has up to three co-examiners
 - No easy way to check accuracy
- Overly positive feedback, minimal critical or constructive feedback
 - Even for known issues
- Varying length & format of feedback
 - Necessitates prompt refinement
 - Need to work through what would be beneficial to examiners
 - Type of information
 - Length
 - Actionable recommendations

Evaluating LLM-Generated Feedback

- Characteristics
 - Does it meet the specifications we requested?
 - Is the correct examiner referred to?
- Quality of the feedback
 - Difficult given the extensive inputs (i.e., transcripts) and lengthy, text-based output
 - How might we do this?
 - Comparison to known issues
 - Expert validation (e.g., Assessment Committee review)
 - Provide to examiners themselves shortly after exam, request their reactions

Ethical Considerations

Ethical Considerations: Short Term

- Where does AI use fit into the human process of exam *delivery* if it is used for exam *scoring*?
- If we know that models and humans each lead to different final pass/fail decisions, which is the "correct" decision? How do we know, and how do we vet these decisions?
- What is the proper role of an AI scorer? Adjudication for examiner disagreements? Quality control as examiners fatigue? Checks on fairness based on candidate demographics?
- If varying the prompt can lead to differences in scores, how can we select the "correct" prompt? (E.g., changes to exemplar case numbers)

Ethical Considerations: Long Term

- How will AI raters impact the desirability of volunteering for the ABS?
- If AI use is adopted, what happens to candidate perceptions of test fairness/quality?
- What degree of ongoing quality control is necessary (and possible) as models change?
- Can a model be allowed to learn from one exam administration to the next?

Open Questions and Future Considerations